Exercise 10 - Inheritance

The final exercise sheet will be something I would say you should be able to solve and write down in about 45 minutes in the exam (If I would be the one giving you the tasks that is), maybe even a bit less. Since this is the first time working with inheritance, you might take a little longer in this exercise tho.

If you have any questions or problems, please write in the Moodle!

Exercise 1 — Persons & Students - Inheritance

For both Exercises create your own class- and header-files. Try out your code in `main.cpp`.

a) Implement a `Class Person`. Each person should have a Name, which is given to them when created, but it cannot be changed after that, yet you should be able to return the `name` of a `Person`.

b) Implement a `Class Student` which inherits from `Person`. Additionally to a name, each Student shall have an `Amount of Semesters` as well as a `matrikelnumber`. Both of these should behave exactly like the `name`, so they are given on creation and afterwards you should only be able to return their values, not change them.

c) For students write a method `incrementSemester`, increasing the `Amount of Semesters` field by exactly one. Finally write the method `print`, which prints out the values of each field of a student onto the console.

Exercise 2 — Methods on Vectors & Data

a) Implement a `Class Data` *which stores a* `Vector` *of* `Students`*. Also implement methods to add a given* `Student`*, remove a* `Student` *at a given* `Index` *and implement a* `print` *method which calls* `student.print()` *on each* `Student` *of the* `Vector`*.*

b) Finally we want to implement two more methods. The first one is called `applyNewSemester` *and increments the* `Number of Semester` *field of every student, while the second one is called* `studiesLessThan` *and shall return a* `Vector` *of all students, who's* `Number of Semester` *is less than a given threshold.*

Important Information for next week:

This is the final exercise sheet, which will be discussed next week. I will also upload my solution for the sheet next week, so you can look it up again later.
There will also be some kind of `checklist` *for you available next week for the kind of tasks I would expect you to be able to solve in the Exam.*